Classical Mechanics

Lecture 5: Pendulums, Oscillators, and Phase Space

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Chapter 1

Pendulums, Oscillators, and Phase Space

We shall work through three examples. The first is simple, the second is much more complicated, and the third is simpler again. There is a reason for this order. The simple and double pendulums show why Lagrange's equations are often far easier to use than direct force resolution. The harmonic oscillator then introduces the variables and geometry of Hamiltonian mechanics.

The advantage is elementary but decisive. Newton's equation asks for accelerations, which are second derivatives. In curvilinear or angular coordinates, their Cartesian components can become unpleasant. A Lagrangian calculation ordinarily asks first for velocities, which are only first derivatives. Once the kinetic and potential energies are known,

$$\mathcal{L} = T - U,$$

the Euler–Lagrange equations turn the remaining work into a fixed procedure. If the resulting differential equations resist analytic solution, they can be integrated numerically. The difficulty of solving a trajectory is separate from the difficulty of formulating the correct equations.

1.1 The Simple Pendulum

Take a mass m at the end of a rigid, massless rod of fixed length r. The rod pivots in a vertical plane. Its only changing coordinate is the angle θ, measured from the downward vertical. With horizontal coordinate x and downward-positive vertical coordinate y,

$$x = r\sin\theta, \qquad y = r\cos\theta. \tag{1.1}$$

Differentiation gives

$$v_x = r\cos\theta\,\dot{\theta}, \qquad v_y = -r\sin\theta\,\dot{\theta}. \tag{1.2}$$

The two trigonometric factors disappear when the speed is squared:

$$\begin{aligned} v^2 &= v_x^2 + v_y^2 \\ &= r^2\dot{\theta}^2\left(\cos^2\theta + \sin^2\theta\right) = r^2\dot{\theta}^2. \end{aligned} \tag{1.3}$$

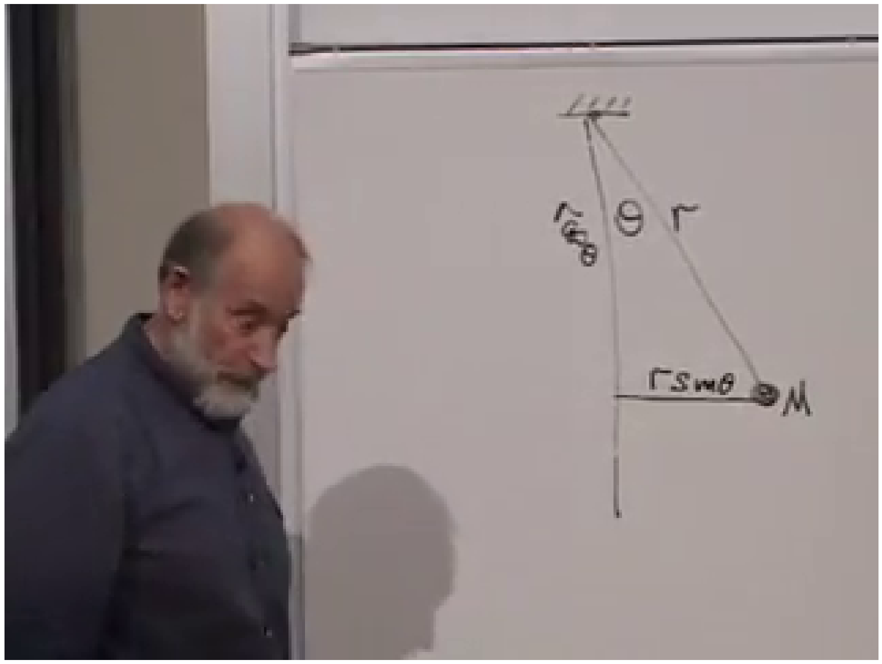

Figure 1.1: The pendulum geometry at 00:05:32. The rod length r, angle θ, and horizontal projection $r\sin\theta$ are complete and unobstructed.

Hence

$$T = \frac{1}{2}mr^2\dot{\theta}^2. \tag{1.4}$$

Only differences of gravitational potential matter. Taking zero potential when the rod is horizontal, the bob's height relative to the pivot is $-r\cos\theta$, so

$$U = -mgr\cos\theta. \tag{1.5}$$

At $\theta = 0$ the bob is at the bottom and $U = -mgr$; at $\theta = \pi$ it is upright and $U = +mgr$.

The Lagrangian is therefore

$$\boxed{\mathcal{L} = \frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta.} \tag{1.6}$$

Question from the audience. Why does the Lagrangian contain $+mgr\cos\theta$ when gravity pulls downward?

Response. The potential itself is $U = -mgr\cos\theta$ in the chosen convention, and the Lagrangian is $\mathcal{L} = T - U$. Subtracting this negative potential produces the plus sign. The force will acquire the expected restoring sign when $\mathcal{L}$ is differentiated with respect to θ.

1.1.1 Equation of motion

The momentum conjugate to θ is

$$\pi_\theta = \frac{\partial\mathcal{L}}{\partial\dot{\theta}} = mr^2\dot{\theta}. \tag{1.7}$$

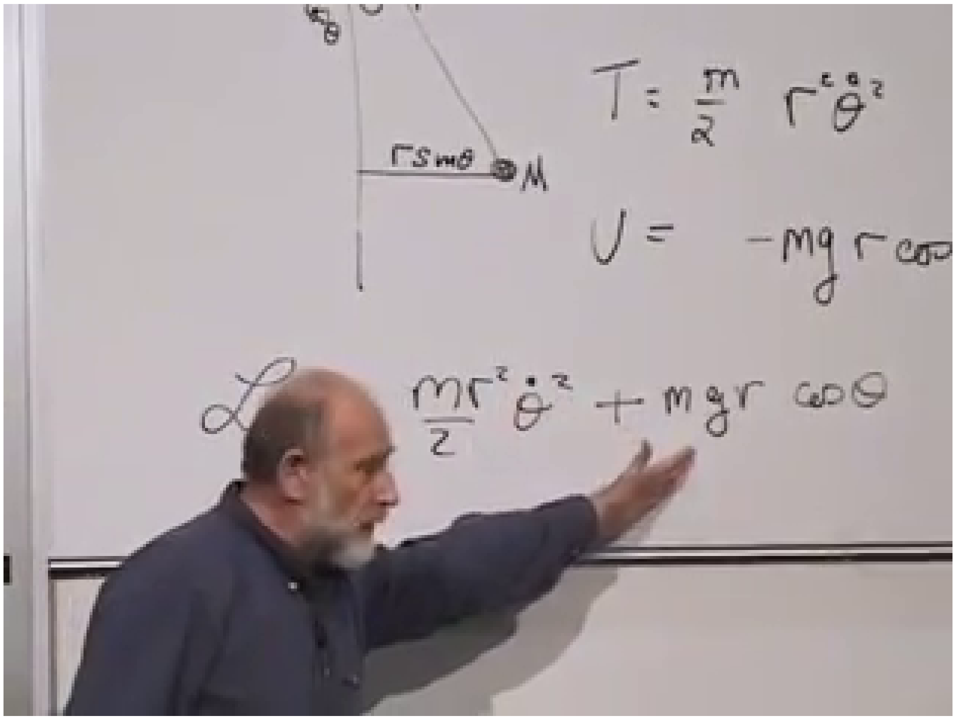

Figure 1.2: The completed kinetic energy, potential energy, and pendulum Lagrangian at 00:10:50. The gesture emphasizes the plus sign created by subtracting the negative gravitational potential.

It is the angular momentum about the pivot, although the derivation needs only its definition. The Euler–Lagrange equation gives

$$\frac{d}{dt}\left(mr^2\dot\theta\right) = -mgr\sin\theta, \tag{1.8}$$

or

$$\boxed{\ddot\theta + \frac{g}{r}\sin\theta = 0.} \tag{1.9}$$

This nonlinear equation can be integrated directly on a computer by advancing θ and $\dot\theta$ over short time steps.

1.1.2 Hamiltonian and energy

There is one generalized coordinate, so the Hamiltonian is

$$\begin{aligned} H &= \pi_\theta\dot\theta - \mathcal{L} \\ &= mr^2\dot\theta^2 - \left(\frac{1}{2}mr^2\dot\theta^2 + mgr\cos\theta\right) \\ &= \frac{1}{2}mr^2\dot\theta^2 - mgr\cos\theta. \end{aligned} \tag{1.10}$$

It is exactly $T + U$, as expected.

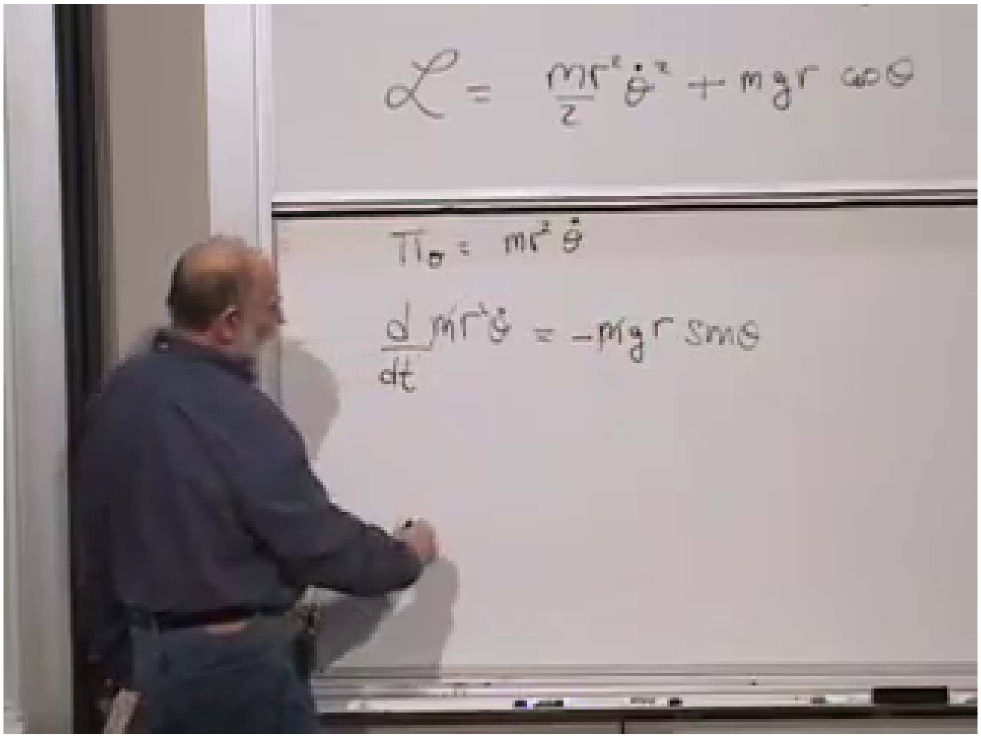

Figure 1.3: The completed pendulum Euler–Lagrange equation at 00:12:27. The upper board retains the Lagrangian from which the lower equation follows.

Question from the audience. Should $\sum_i \pi_i \dot{q}_i$ always equal $2T$?

Response. It does for the ordinary kinetic energies used here. If T is homogeneous of degree two in the generalized velocities, Euler's homogeneous-function theorem gives $\sum_i(\partial T/\partial \dot{q}_i)\dot{q}_i = 2T$. It is not a universal identity for every possible Lagrangian, so it should be checked rather than assumed.

1.2 The Pendulum Energy Landscape

We do not need the exact nonlinear solution to classify the motion. The potential changes by

$$U(\pi) - U(0) = mgr - (-mgr) = 2mgr \tag{1.11}$$

between the bottom and the upright position.

Suppose the pendulum starts at the bottom with a push. If its initial kinetic energy is less than $2mgr$, it spends that energy climbing, reaches a turning point with $\dot{\theta} = 0$, and returns. It oscillates between two turning points. If the kinetic energy exceeds $2mgr$, some remains when the bob passes the upright point, and the pendulum continues to rotate. The rotation may be clockwise or counterclockwise. Exactly at the threshold, the ideal trajectory approaches the unstable upright configuration with zero speed. This separatrix motion is possible but requires exact tuning.

Question from the audience. What is meant by the *top* of the orbit? Is it the ceiling?

Response. The pivot is imagined to permit a complete 360° rotation. The top is the configuration $\theta = \pi$, with the bob vertically above the pivot. The phrase does not refer to a physical ceiling obstructing the rod.

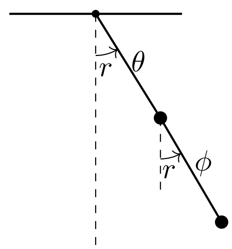

Figure 1.4: Both angles are measured from the same vertical direction. This choice makes a common rotation appear as equal shifts of θ and ϕ.

Thus the familiar swinging pendulum and the continuously rotating pendulum are different families of motion separated by the threshold energy. This qualitative conclusion follows immediately from conservation of Eq. (1.10).

1.3 The Double Pendulum

The single pendulum is simple enough that direct force resolution remains possible. The double pendulum is the real test. Its motion can become chaotic, and resolving all rod tensions and accelerations by $F = ma$ is treacherous. The Lagrangian method, by contrast, changes only the amount of algebra.

Take two equal point masses m joined by two massless rods of equal length r, moving in one vertical plane. Let θ be the angle of the first rod from the downward vertical and let ϕ be the angle of the second rod from the same vertical.

Question from the audience. Must the double pendulum remain in one plane?

Response. No. A spatial double pendulum is a legitimate problem, but it needs more coordinates. We choose planar motion here so that the two angles θ and ϕ form a minimal complete set.

The first mass has kinetic energy

$$T_1 = \frac{1}{2} m r^2 \dot{\theta}^2. \tag{1.12}$$

The second mass moves both because the first mass moves and because the second rod rotates relative to the first mass. Its velocity components are

$$v_{2x} = r\cos\theta\,\dot{\theta} + r\cos\phi\,\dot{\phi}, \tag{1.13}$$

$$v_{2y} = -r\sin\theta\,\dot{\theta} - r\sin\phi\,\dot{\phi}. \tag{1.14}$$

Squaring and adding produces two diagonal terms and one mixed term:

$$\begin{aligned} v_2^2 &= r^2\dot{\theta}^2 + r^2\dot{\phi}^2 \\ &\quad + 2r^2\dot{\theta}\dot{\phi}\left(\cos\theta\cos\phi + \sin\theta\sin\phi\right). \end{aligned} \tag{1.15}$$

The trigonometric identity

$$\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi) \tag{1.16}$$

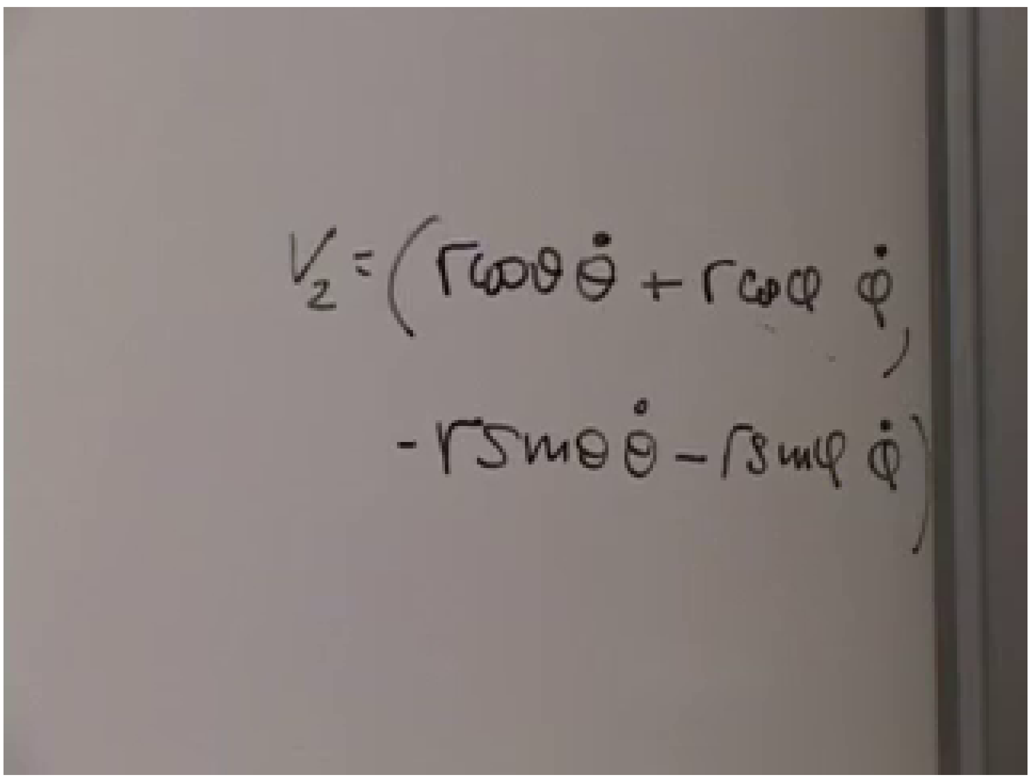

Figure 1.5: The two contributions to the lower bob's velocity at 00:29:02. Each Cartesian component contains motion inherited from the upper bob and motion due to the second rod.

then gives

$$T_2 = \frac{1}{2}mr^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi). \tag{1.17}$$

Adding the first bob,

$$\boxed{T = mr^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi).} \tag{1.18}$$

Question from the audience. Why did the factor 1/2 disappear from the $\dot{\theta}^2$ term?

Response. Both masses carry the motion generated by the first rod. The upper bob contributes $\frac{1}{2}mr^2\dot{\theta}^2$, and the lower bob contributes another identical term. Their sum is $mr^2\dot{\theta}^2$. The simple coefficient depends on our choice of equal masses and equal rod lengths.

The gravitational energy must also count the first rod twice, because both masses rise when θ changes:

$$\begin{aligned} U &= -mgr\cos\theta - mg(r\cos\theta + r\cos\phi) \\ &= -2mgr\cos\theta - mgr\cos\phi. \end{aligned} \tag{1.19}$$

Question from the audience. Are the two gravitational energies measured from the same reference point?

Response. Yes. The lower bob's vertical coordinate is the sum of the two rod projections. Its energy therefore includes both $-mgr\cos\theta$ and $-mgr\cos\phi$, while the upper bob supplies a second $-mgr\cos\theta$. A common additive constant would still be irrelevant.

The full Lagrangian is

$$\boxed{\begin{aligned} \mathcal{L} = mr^2\dot{\theta}^2 &+ \frac{1}{2}mr^2\dot{\phi}^2 \\ &+ mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi) \\ &+ mgr(2\cos\theta + \cos\phi). \end{aligned}} \tag{1.20}$$

Nothing was guessed. We chose coordinates, differentiated positions, formed T and U, and subtracted.

1.4 Rotation Symmetry Without Gravity

Gravity singles out the vertical direction, so the full Lagrangian Eq. (1.20) is not invariant under a common planar rotation. Now imagine the apparatus far from a gravitational field. Removing the last two potential terms leaves

$$\mathcal{L}_0 = mr^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi). \tag{1.21}$$

It is invariant under

$$\theta \longmapsto \theta + \epsilon, \qquad \phi \longmapsto \phi + \epsilon. \tag{1.22}$$

Adding a constant changes neither angular velocity, and the difference $\theta - \phi$ remains fixed.

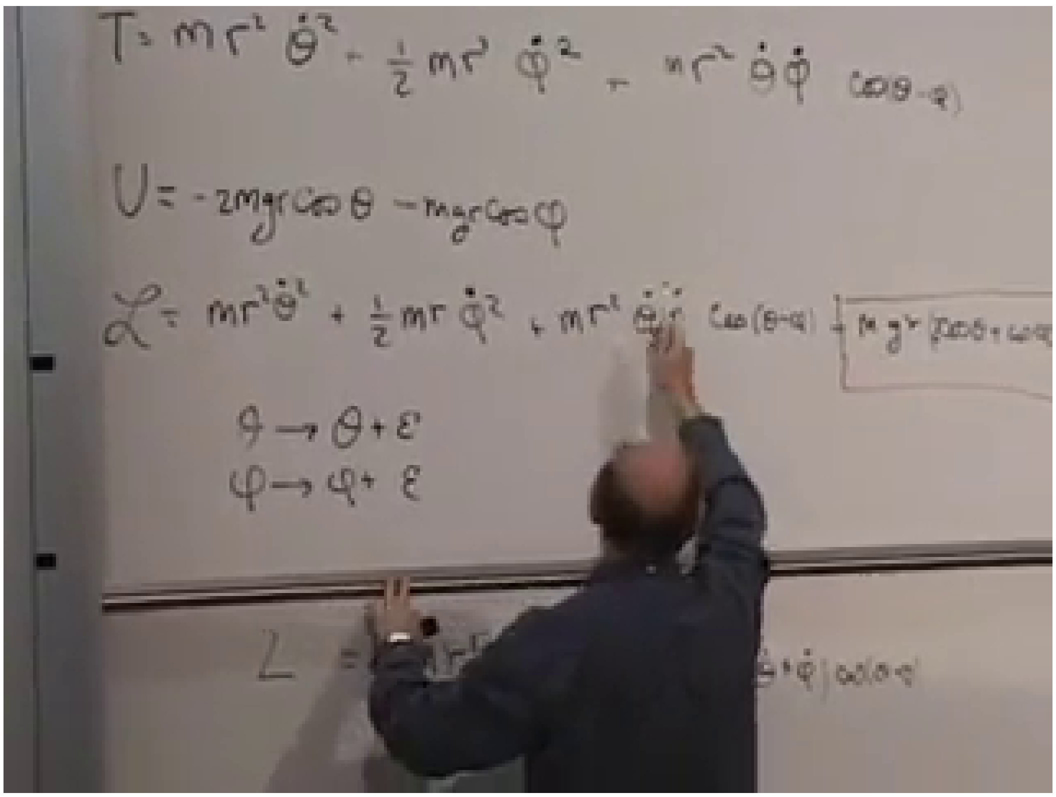

Figure 1.6: The double-pendulum kinetic energy, potential energy, Lagrangian, and common angle shift at 00:52:47. The board preserves the $\cos(\theta - \phi)$ coupling that controls the later derivatives.

Question from the audience. Why does the coupling contain only $\theta - \phi$, rather than $\theta + \phi$?

Response. Without gravity there is no absolute direction in the plane. A common rotation must leave the kinetic energy unchanged. $\theta - \phi$ is invariant under that rotation, whereas $\theta + \phi$ changes by 2ϵ. The trigonometric identity in Eq. (1.16) is therefore also a visible statement of rotational symmetry.

For this symmetry the two generators are both 1. Noether's charge is

$$Q = \pi_\theta + \pi_\phi. \tag{1.23}$$

The mixed kinetic term makes each canonical momentum depend on both angular velocities:

$$\pi_\theta = 2mr^2\dot\theta + mr^2\dot\phi\cos(\theta - \phi), \tag{1.24}$$
$$\pi_\phi = mr^2\dot\phi + mr^2\dot\theta\cos(\theta - \phi). \tag{1.25}$$

Thus

$$\boxed{\begin{aligned} Q = mr^2\Big[&2\dot\theta + \dot\phi \\ &+ (\dot\theta + \dot\phi)\cos(\theta - \phi)\Big] \\ = \text{constant}. \end{aligned}} \tag{1.26}$$

This is the total angular momentum. Once its initial value is known, one relation among the velocities remains fixed throughout the motion.

1.4.1 The coupled equations

The equations may be unpleasant, but their construction is unambiguous:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot\theta} = \frac{\partial \mathcal{L}}{\partial \theta}, \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot\phi} = \frac{\partial \mathcal{L}}{\partial \phi}. \tag{1.27}$$

Expanding Eq. (1.20) carefully, with $\Delta = \theta - \phi$, gives the useful implicit pair

$$2\ddot\theta + \ddot\phi\cos\Delta + \dot\phi^2\sin\Delta + 2\frac{g}{r}\sin\theta = 0, \tag{1.28}$$
$$\ddot\phi + \ddot\theta\cos\Delta - \dot\theta^2\sin\Delta + \frac{g}{r}\sin\phi = 0. \tag{1.29}$$

Setting $g = 0$ recovers the rotation-invariant case.

Question from the audience. When differentiating $\cos(\theta - \phi)$, where do the opposite signs enter?

Response. A partial derivative with respect to θ gives $-\sin(\theta - \phi)$, while a partial derivative with respect to ϕ gives $+\sin(\theta - \phi)$. A time derivative also differentiates both angles:

$$\frac{d}{dt}\cos(\theta - \phi) = -\sin(\theta - \phi)(\dot\theta - \dot\phi).$$

Keeping partial and total derivatives distinct removes the apparent ambiguity.

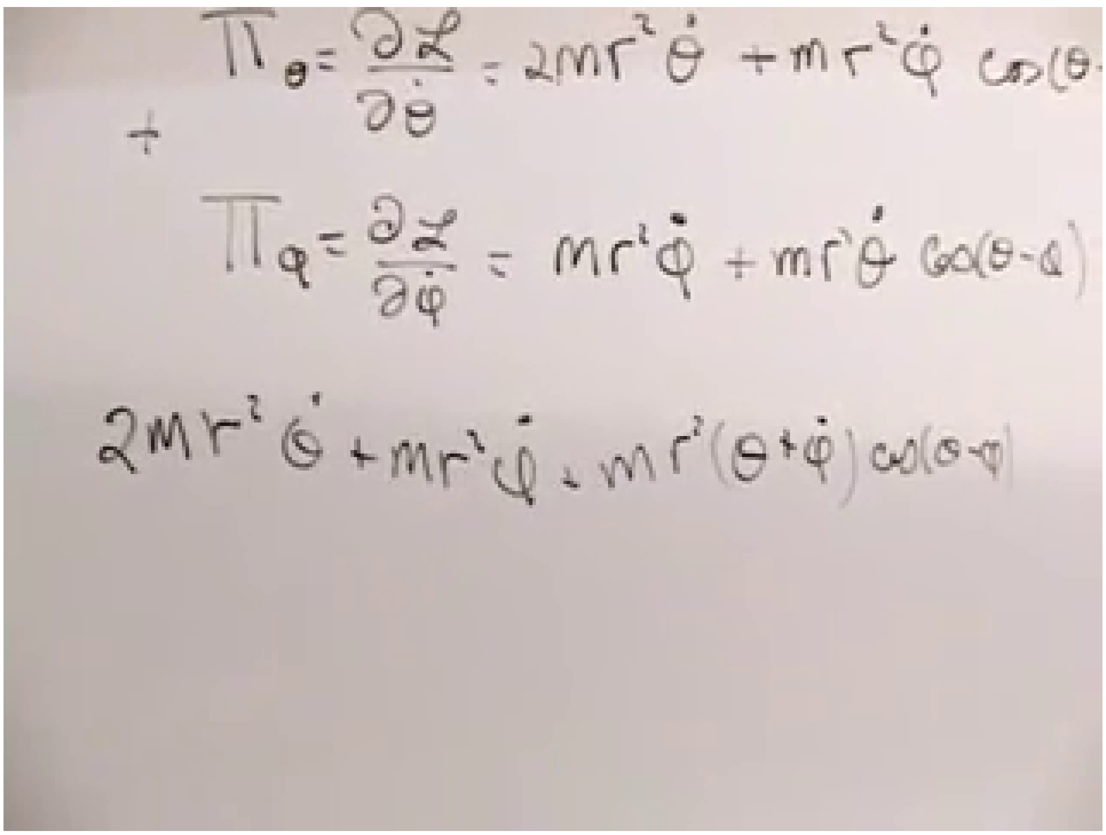

Figure 1.7: The two canonical momenta and their conserved sum at 00:46:15. The mixed velocity dependence is complete and unobstructed.

The energy is obtained by reversing the sign of the potential contribution:

$$
\begin{aligned}
E = T + U = mr^2\dot\theta^2 &+ \frac{1}{2}mr^2\dot\phi^2 \\
&+ mr^2\dot\theta\dot\phi\cos\Delta \\
&- mgr(2\cos\theta + \cos\phi).
\end{aligned} \tag{1.30}
$$

Without gravity, energy and Q provide two conserved relations. They do not make the trajectories visually simple, but they substantially reduce the problem. This is the desired lesson: difficult mechanics can still be reduced to a reliable algorithm.

1.5 Why Harmonic Motion Is Universal

We now turn to the most basic nontrivial problem in theoretical physics. A free particle is simpler, but the harmonic oscillator is the structure that reappears whenever a stable system is disturbed slightly from equilibrium. The pendulum already contains it.

Near $\theta = 0$,

$$
\begin{aligned}
U(\theta) &= -mgr\cos\theta \\
&= -mgr + \frac{1}{2}mgr\,\theta^2 \\
&\quad - \frac{1}{24}mgr\,\theta^4 + \cdots .
\end{aligned} \tag{1.31}
$$

The constant $-mgr$ has no effect on the equations of motion. To quadratic order,

$$
\mathcal{L} \simeq \frac{1}{2}mr^2\dot\theta^2 - \frac{1}{2}mgr\,\theta^2, \tag{1.32}
$$

which is the Lagrangian of a harmonic oscillator.

Question from the audience. What happened to the constant $-mgr$?

Response. It remains in the numerical value assigned to the energy, but it has zero derivative. Adding or subtracting a constant from $\mathcal{L}$ changes neither Euler–Lagrange equation. We may choose the zero of potential for convenience.

The canonical realization is a mass on a spring. Let $x = 0$ be its equilibrium position and write

$$
U(x) = \frac{1}{2}kx^2, \qquad \mathcal{L} = \frac{1}{2}m\dot x^2 - \frac{1}{2}kx^2. \tag{1.33}
$$

The force is

$$
F = -\frac{dU}{dx} = -kx, \tag{1.34}
$$

which is Hooke's law. Positive displacement produces a negative force and negative displacement produces a positive force: both point back toward equilibrium.

This law is not an exact global description of a real spring. Stretch far enough and nonlinearities appear; farther still, the spring breaks. Its generality is local. For a smooth potential expanded about $x = 0$,

$$
U(x) = U_0 + ax + \frac{1}{2}kx^2 + c_3x^3 + c_4x^4 + \cdots . \tag{1.35}
$$

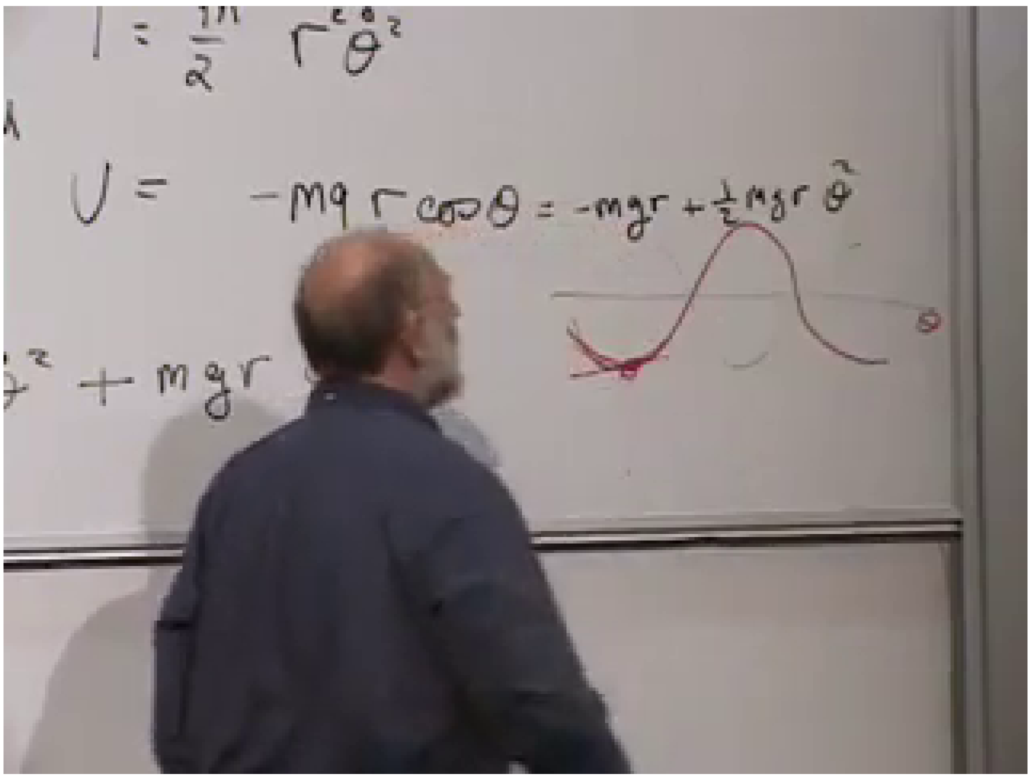

Figure 1.8: The exact pendulum potential and its quadratic approximation at 00:59:15. The parabola agrees with the cosine potential near the stable minimum and departs from it for larger angles.

At an equilibrium $a = U'(0) = 0$. At a stable nondegenerate minimum $k = U''(0) > 0$, so the quadratic term dominates all higher powers for sufficiently small x.

Question from the audience. What if a linear term appears in the expansion?

Response. Then $x = 0$ was not chosen at the equilibrium. Shift the origin to the point where $U'(x) = 0$; the linear term disappears. The constant term fixes only the zero of energy, while the quadratic term controls the leading restoring force.

Question from the audience. Is Hooke's law exact, and can it be derived without experiment?

Response. Its coefficient k and its range of validity must be measured. Mathematics explains why a quadratic approximation is generic near a nondegenerate minimum, not why a particular material has a particular k. The approximation improves as the displacement shrinks, and its error is estimated by comparing the higher terms with $kx^2/2$.

There is one refinement worth making. A cubic correction may accompany a positive quadratic term. But if the quadratic coefficient were tuned to zero and $x = 0$ were still a genuine smooth minimum, a cubic term could not be the leading nonzero term, because an odd power changes sign across the origin. The next stable leading possibility is typically quartic. The absence of the ordinary quadratic term is therefore exceptional.

This same small-oscillation logic applies to mechanical vibration, sound, water waves, molecular motion, and fields. It fails for sufficiently large amplitude, where the higher terms can no longer be neglected.

Question from the audience. If the small-oscillation approximation is initially valid, what prevents the motion from growing until it leaves that regime?

Response. Energy conservation. For the stable oscillator,

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2,$$

and both terms are nonnegative. Small initial energy bounds both $|x|$ and $|\dot{x}|$. An inverted potential has the opposite behavior: near the top of a hill the quadratic potential is negative, and a small displacement can grow rather than oscillate.

1.6 Equation and General Solution

Applying the Euler–Lagrange equation to Eq. (1.33),

$$m\ddot{x} = -kx. \tag{1.36}$$

Define

$$\omega^2 = \frac{k}{m}. \tag{1.37}$$

Then

$$\boxed{\ddot{x} + \omega^2 x = 0.} \tag{1.38}$$

Only the ratio k/m controls the time dependence. Increasing k and m by the same factor leaves the equation unchanged.

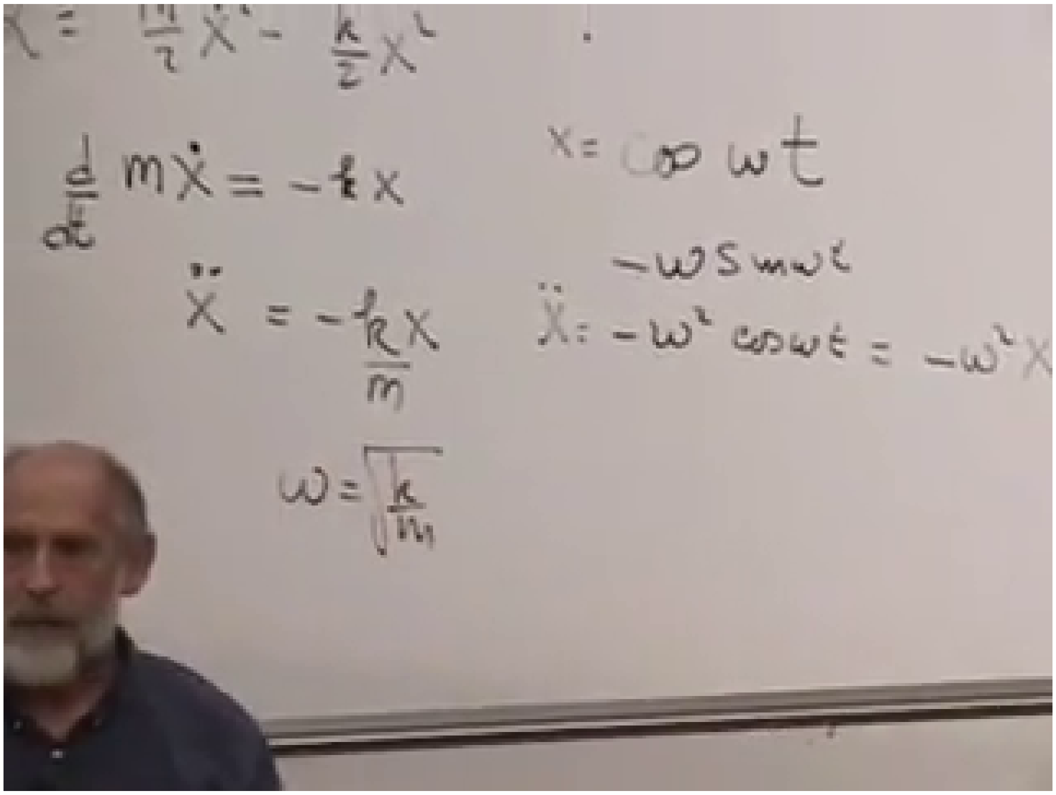

Figure 1.9: The oscillator equation, cosine trial solution, and $\omega = \sqrt{k/m}$ at 01:20:42.

The reason sines and cosines solve the equation is immediate:

$$\frac{d^2}{dt^2} \cos \omega t = -\omega^2 \cos \omega t,$$
$$\frac{d^2}{dt^2} \sin \omega t = -\omega^2 \sin \omega t.$$

The general real solution is

$$x(t) = A \cos \omega t + B \sin \omega t. \tag{1.39}$$

Two constants are required because the equation is second order: initial position and initial velocity must both be specified. The same family may be written

$$x(t) = C \cos[\omega(t - t_0)] \,, \tag{1.40}$$

where C is the amplitude and t_0 specifies the phase, here chosen as a time at which x is maximal.

Question from the audience. Is the phase $t - t_0$, t_0, or $\omega(t - t_0)$?

Response. The dimensionless phase appearing inside the cosine is $\omega(t - t_0)$. The parameter t_0 is a time shift, often loosely called the phase parameter. Equivalently one may write $C \cos(\omega t - \delta)$ with dimensionless phase $\delta = \omega t_0$.

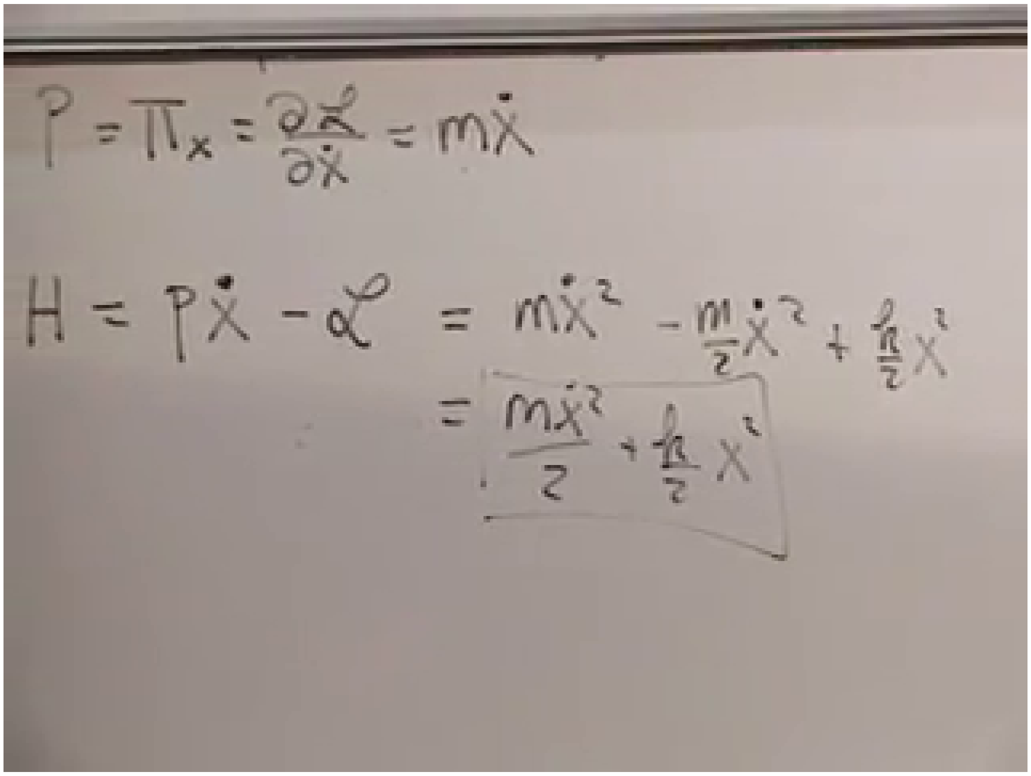

Figure 1.10: The canonical momentum and Hamiltonian calculation at 01:26:53. The completed board state is visible without the earlier hand occlusion.

1.7 From Velocities to Canonical Momenta

For the oscillator,

$$p = \pi_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}. \tag{1.41}$$

The Hamiltonian is

$$\begin{aligned} H &= p\dot{x} - \mathcal{L} \\ &= m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2\right) \\ &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \end{aligned} \tag{1.42}$$

Hamiltonian mechanics takes x and p, rather than x and $\dot{x}$, as its fundamental variables. Here the conversion is immediate:

$$\dot{x} = \frac{p}{m}. \tag{1.43}$$

Substitution gives

$$\boxed{H(x,p) = \frac{p^2}{2m} + \frac{1}{2}kx^2.} \tag{1.44}$$

Notice the useful change: m multiplies $\dot{x}^2/2$, but it divides $p^2/2$. This is a mnemonic for ordinary quadratic kinetic energy, not a substitute for carrying out the Legendre transform.

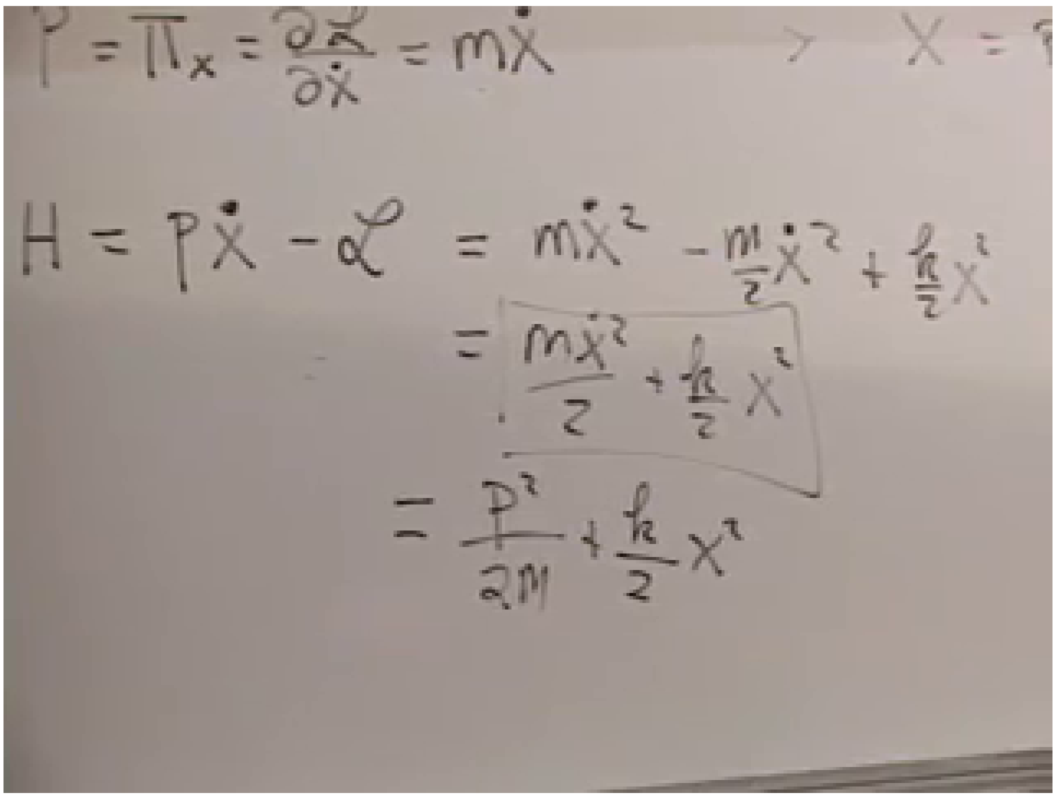

Figure 1.11: The oscillator Hamiltonian rewritten in canonical variables at 01:33:32. The final line contains $p^2/(2m) + kx^2/2$.

Question from the audience. Why reorganize mechanics around q and p instead of q and $\dot{q}$?

Response. The full motivation is not yet visible. Hamilton's equations will place coordinates and momenta in a nearly symmetric first-order form, and this structure becomes indispensable in quantum mechanics. For now the oscillator provides the first hint: after a rescaling, H treats the quadratic coordinate and momentum terms symmetrically.

Question from the audience. Does the oscillator have a special role in quantum mechanics too?

Response. Yes. Small disturbances about equilibrium lead to oscillators in quantum theory for the same local reason. Quantum mechanics adds zero-point energy, so the energy and fluctuation amplitude cannot both be reduced arbitrarily to zero. That belongs to the quantum course; here we retain the classical limit.

1.8 Motion in Phase Space

The plane with horizontal coordinate x and vertical coordinate p is called phase space. A point (x, p) specifies the instantaneous state of this one-dimensional system. As time passes, that point traces a curve.

Energy conservation confines the curve to

$$\frac{p^2}{2m} + \frac{1}{2}kx^2 = E. \tag{1.45}$$

This is an ellipse. Its intercepts are

$$x_{\max} = \sqrt{\frac{2E}{k}}, \qquad p_{\max} = \sqrt{2mE}. \tag{1.46}$$

Changing E changes the size but not the eccentricity of the ellipses. Rescaling the axes by

$$X = \sqrt{k}\,x, \qquad P = \frac{p}{\sqrt{m}}$$

turns them into circles $X^2 + P^2 = 2E$.

The phase point goes once around its ellipse in the oscillator period

$$\tau = \frac{2\pi}{\omega}. \tag{1.47}$$

Every ellipse has the same period because the ideal oscillator frequency is independent of amplitude. In the rescaled plane the entire family rotates uniformly with angular frequency ω.

Ordinary oscillation is a projection of this motion. Projection onto the x-axis gives the back-and-forth displacement; projection onto the p-axis gives the momentum oscillation, shifted by a quarter period.

Question from the audience. Is the oscillator moving fastest when it is farthest from the origin?

Response. No. At $x = \pm x_{\max}$, the phase ellipse meets the x-axis, so $p = 0$ and the mass is momentarily at rest. At $x = 0$, the ellipse reaches $p = \pm p_{\max}$, so the speed is greatest.

The ellipse is special to the harmonic oscillator, but the larger statement is general: for a time-independent Hamiltonian, trajectories remain on constant-energy sets in phase space.

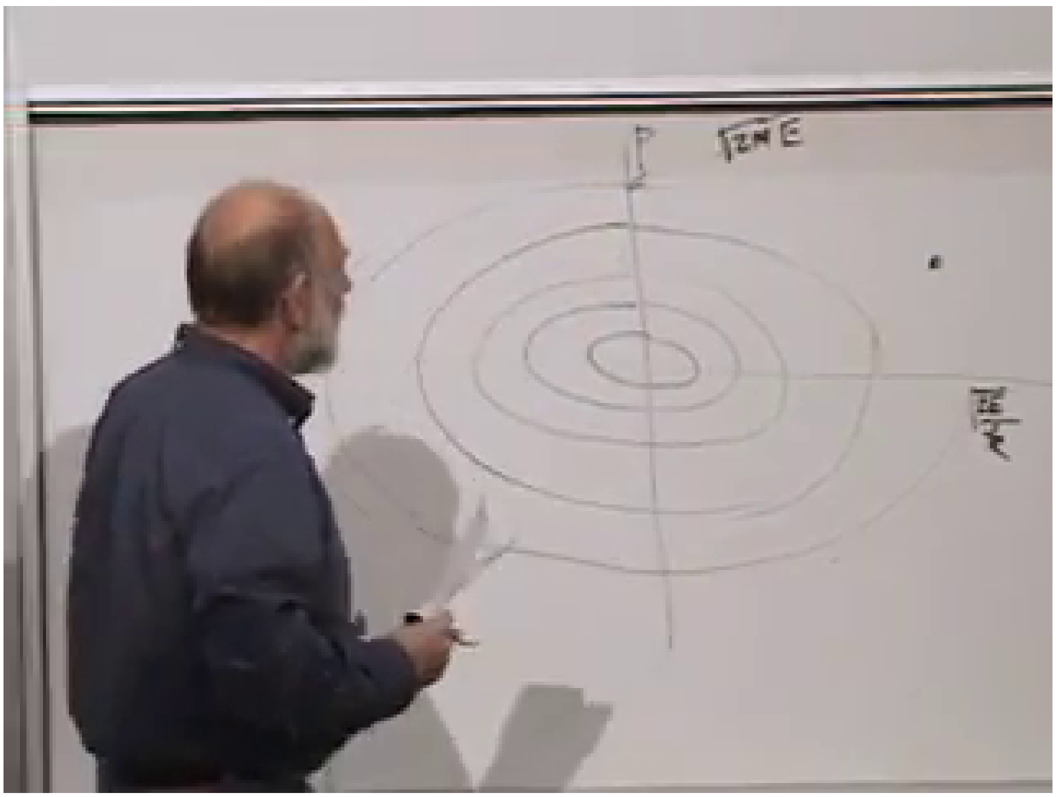

Figure 1.12: Nested constant-energy ellipses in the x-p plane at 01:40:34. The board also marks the momentum and position intercepts.

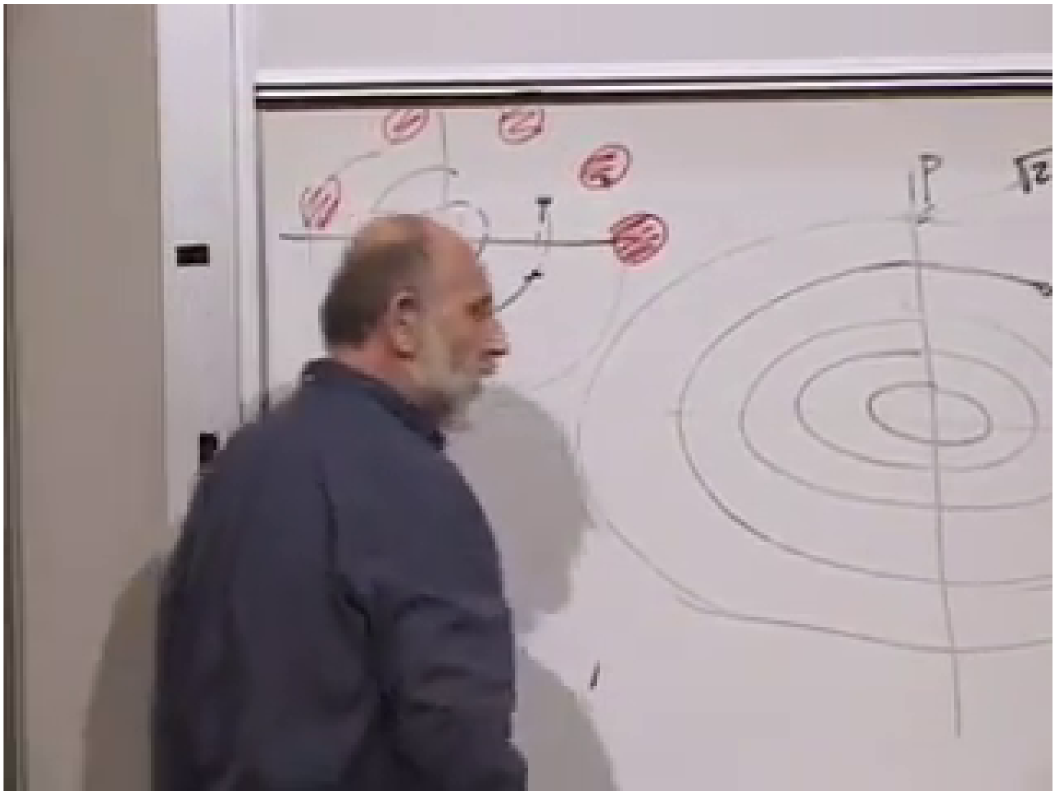

Figure 1.13: A patch of neighboring initial conditions transported around the phase portrait at 01:45:51. Its position changes while its phase-space area is preserved.

1.8.1 Area and invertibility

Imagine a small patch of initial conditions rather than one perfectly known point. Oscillator evolution carries the patch around phase space without changing its area. The corresponding general theorem is Liouville's theorem: Hamiltonian flow preserves phase-space volume. Its proof belongs to the Hamiltonian development that follows.

One technical condition is already unavoidable. To replace velocities by momenta, the relations

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \tag{1.48}$$

must be invertible for $\dot{q}_i(q, p)$. For an ordinary unconstrained system, local invertibility requires

$$\det\left(\frac{\partial^2 \mathcal{L}}{\partial \dot{q}_i\,\partial \dot{q}_j}\right) \neq 0. \tag{1.49}$$

Question from the audience. What happens if one momentum corresponds to several velocities, or the relation cannot be inverted?

Response. Then the ordinary Legendre transform does not define a single Hamiltonian in the variables q, p. For elementary unconstrained mechanics that signals a defective formulation. More generally, singular Lagrangians can describe constrained or gauge systems, but they require a separate constrained-Hamiltonian treatment.

We have therefore arrived at the threshold of Hamiltonian mechanics. The examples began as a practical argument for calculating velocities instead of accelerations. They end with a new state space, a conserved function $H(q, p)$, motion along its level sets, and an area-preserving flow. Those structures, rather than any one pendulum trajectory, are what we shall develop next.